TO: Head of Fraud Risk, Deacon Financial Services

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SUBJECT: Recommended fraud review model and cut-off for new account applications

1. Executive Summary

Our team combined the HW3 feature ideas from all five members into a 20-variable logistic regression that scores each application when it is submitted, and on held-out test applications it reached an AUC of 0.844, compared with 0.651 for the bank's current fraud score. We recommend sending any application with a predicted fraud probability above 0.12 to manual review, which would mean reviewing about 7% of applications and catching 48% of fraud. On the test set, this cuts fraud-related costs from $1.03M to $0.71M (about 31%), and the current review teams have room for the added cases.

2. Objective

When an application is submitted, DFS must decide whether to approve it or send it to manual review. We were asked to build a logistic regression that estimates the chance an application turns out to be confirmed fraud and to recommend where the review cut-off should be.

This decision has become more important since July 1, 2026. The New Money Bonus is the first offer in the data that pays a $200 bonus, and monthly applications grew about four times after it launched. Bonus abuse and synthetic identity fraud grew with it.

Both types of errors are costly. When a fraudulent application is approved, the bank pays out the bonus and referral reward and later spends time investigating. When a legitimate customer is sent to review, the bank pays for the review, and some of those customers drop out before finishing. The goal is a cut-off that catches more fraud without delaying too many legitimate customers.

3. Key Findings

- Our five HW3 memos produced 55 candidate features. Several team members proposed similar ideas (three of us proposed a shared-device count, and others proposed different versions of email risk and session speed), so we kept one version of each. After screening and variable selection, 20 variables remained: 9 identity and email, 6 application behavior, 2 device, 2 referral, and 1 missing-data flag. We could not build the shared phone, address, and payout-account features because those tables are empty for our login.
- The model separates fraud from good applications much better than the bank's current score (test AUC 0.844 vs. 0.651; Somers' D 0.69). If the bank reviewed only the riskiest 5% of applications, it would find 40.6% of all fraud with our model compared with 6.4% with the current score. Training AUC was 0.849, very close to the test AUC, so we do not see signs of overfitting.
- At a 0.12 cut-off, the model catches 48% of fraud and flags 7% of applications. About one in three flagged applications is confirmed fraud, and 95% of good applicants are approved without review. It does best on bonus abuse (68% caught) and worst on first-party fraud (22%).

- The strongest predictor is how many times the same person has applied before. Each time that number doubles, the odds of fraud roughly triple (likelihood ratio test $p < 0.001$). The next strongest are a session that runs slower than normal for its channel, a device the bank has seen before, and a disposable email domain, which raises the odds by about 1.8 times.

4. Implications & Recommendations

1. Send applications scoring above 0.12 to manual review. The ROC curve shows the trade-off at every cut-off but does not choose one, so we picked the cut-off with the lowest total cost. A missed fraud costs $57 in onboarding and verification and an estimated 4 hours of investigation ($248), plus the $200 bonus if the person applied after the bonus launched and the referral reward if a referral code was used. That comes to $305 to $555, about $400 on average. Reviewing a good customer costs about $80 (the $48 review plus value lost from the roughly 16% who drop out). Because a missed fraud costs about five times more than a review, the cut-off should be well below 0.5. On the test set, cost falls from $1.03M with no model to $0.71M. A 0.5 cut-off would miss 87% of fraud, and the Youden cut-off would send one in five applicants to review.
2. The cut-off still saves money with lower cost estimates. If we count only the bonus (when it applies) and onboarding costs for a missed fraud and $48 per review, the lowest-cost cut-off becomes 0.20, and test cost drops from $344K to $267K. Our 0.12 cut-off still costs less than having no model ($288K).
3. The review teams have capacity for the added volume. Applied to all August 2026 applications, the cut-off would flag about 3,900. In August the three fraud review teams had about 21,500 cases of scheduled capacity and received 3,300, so there is room for about 18,000 more.
4. Retrain the model every month while the bonus is running. Fraud went from 4.7% of applications (January 2025 to July 2026) to 5.9% in August, and bonus abuse grew the fastest. We suggest tracking AUC, the share of fraud caught, and the share of flagged applications that turn out to be fraud each month.
5. Work on the fraud types the model misses. Stolen identity and first-party fraud are the hardest to catch at application time. Getting access to the shared phone, address, and payout-account data, and adding a second check 30 days after the account opens, would likely help the most.

The analysis has several limitations. The 4 hours of investigation per case is our estimate because case hours are not recorded, and money stolen after the account opened is not in our data. Some fraud is also missing from the labels, since recent investigations are still open and declined applications are rarely investigated. Finally, investigators often open cases because of repeat applications or shared devices, which may make those predictors appear somewhat stronger than they are.

5. Supporting Analysis

We built one row per application from the DFS warehouse. An application counts as fraud if an investigation confirmed it. We left out September 2026 because most of its investigations were still open. Following the course code, we split the data 75% training and 25% testing at random.

We made sure every feature only used information the bank would have when the application came in.

For data preparation, we filled missing values with the median and added a flag for whether the value was missing, since missing device and session data were related to fraud. We log-transformed skewed counts, dropped features that were mostly missing or nearly identical to another one, and screened the rest with chi-square and F-tests. We then ran forward, stepwise, and backward selection with 5-fold cross-validation. We used stepwise, and all three picked the same 20 variables. Every variable had a VIF under 10, so multicollinearity was not a concern.

For costs, a missed fraud only includes the $200 bonus if the bonus was running when the person applied, and the referral reward only if they used a referral code. Most of the fraud in the data happened before the bonus started, so charging the full amount for every case would have overstated the losses.

6. Appendix

Table A1. Model performance on the test set (52,998 applications, 4.9% fraud)

Model	Variables	Train AUC	Test AUC	Fraud caught at its cut-off	Flagged	Test-set cost
Bank's current fraud score	n/a	n/a	0.651	n/a	n/a	n/a
First-pass model (initial feature set)	8	0.783	0.775	37.1%	5.1%	$751,949
Final team model	20	0.849	0.844	48.3%	7.2%	$713,979

No model (approve everything): $1,032,428. Costs use $305 to $555 per missed fraud, depending on the offer and referral code, and $80 per reviewed good applicant.

Table A2. Cut-off options for the final model, test set

Cut-off rule	Cut-off	Flagged	Fraud caught	Precision	Total cost
Default	0.50	1.3%	12.8%	47.2%	$907,867
Youden's J	0.053	18.8%	70.5%	18.3%	$944,196
Max F1	0.14	6.0%	44.6%	36.4%	$710,054
Minimum cost (recommended)	0.12	7.2%	48.3%	33.0%	$713,979

Cut-offs were chosen on the training set and applied to the test set. The F1 cut-off happens to cost about the same on the test set (within 1%). We kept the cost-based cut-off because it was chosen using costs.

Table A3. Lift, test set

Review the riskiest	Share flagged that is fraud	Lift vs. random	Share of all fraud found
1%	48.0%	9.8x	9.8%
5%	39.7%	8.1x	40.6%
10%	27.3%	5.6x	55.9%
20%	17.5%	3.6x	71.6%

Table A4. Final model: top drivers (ranked by standardized effect)

Variable	Category	Odds ratio	Reading
Prior applications, same applicant (log)	Application behavior	5.06	Doubling prior applications triples the odds
Session length vs. channel median (log)	Application behavior	1.55	Slower than others on the same channel: higher risk
Device age, days known (log)	Device linkage	1.39	Older, reused devices: higher risk
Disposable email domain	Identity & email	1.76	Raises odds 76%
Email risk score (0–4)	Identity & email	1.30	Each point raises odds 30%
Referral code burst (log)	Incentive & referral	2.41	Doubling recent code use raises odds 1.8 times
DOB-linked emails in 4 weeks	Identity & email	1.12	Each extra email raises odds 12%
Seconds per page (log)	Application behavior	0.75	Faster per page: higher risk
Branch application, home zip different from branch zip	Application behavior	0.71	Lower risk after other factors
Used a referral code	Incentive & referral	1.59	Raises odds 59%
Other applicants on same device, 90 days (log)	Device linkage	1.16	Doubling raises odds 11%
Valid home phone	Identity & email	0.79	Lowers odds 21%

All 12 are significant at $p < 0.001$. Three of the 20 variables (email domain surge, $p = 0.19$; session z-score, $p = 0.24$; no valid phone, $p = 0.74$) were kept by stepwise selection because they improved cross-validated AUC, but they are not significant on their own. Full output is in the Jupyter notebook.